

# Jennifer Brana

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## Research Interests

I am interested in improving the efficiency and programmability of computing systems. I aim to improve how compilers and systems reason about performance to enable efficient execution across diverse architectures.

*Research areas:* performance modeling & analysis; compilers; computer systems; computer architecture; ML systems.

## Education

### Carnegie Mellon University

MS IN COMPUTER SCIENCE RESEARCH

Pittsburgh, PA

June 2023 - May 2026

Coursework in compilers, computer architecture, deep learning systems, automated reasoning, operating systems, and secure systems.

### University of Portland

B.S. IN COMPUTER SCIENCE, *Cum Laude*

Portland, OR

Aug. 2019 - May 2023

MINOR IN COMPUTER ENGINEERING.

## Refereed Journal Publications

### Kobold: Simplified Cache Coherence for Cache-Attached Accelerators

IEEE CAL 2023

Jennifer Brana, Brian C. Schwedock, Yatin A. Manerkar, Nathan Beckmann

## Refereed Workshop Presentations

### Harmony: Co-Optimizing Parallelism and Locality to Bound Performance

YARCH 2025

Jennifer Brana, Nathan Beckmann

### Kobold: Simplified Cache Coherence for Cache-Attached Accelerators

Student Research Competition @

MICRO 2022

Jennifer Brana, Brian C. Schwedock, Yatin A. Manerkar, Nathan Beckmann

### Kobold: Simplified Cache Coherence for Cache-Attached Accelerators

WDDSA @ MICRO 2022

Jennifer Brana, Brian C. Schwedock, Yatin A. Manerkar, Nathan Beckmann

## Honors & Awards

2023 **NSF Graduate Research Fellowship**, National Science Foundation

2023 **Cum Laude**, University of Portland

2023 **Outstanding Student in Computer Science**, University of Portland

2020 **Tau Beta Pi Induction**, Oregon Gamma, University of Portland

All terms **Dean's List**, University of Portland

2019-2023 **President's Scholarship**, University of Portland

2019-2023 **FIRST Robotics Scholarship**, University of Portland

## Research Experience

### Unaffiliated

INDEPENDENT RESEARCHER

August 2026 - Present

- Characterized performance bottlenecks in SAT solvers. Developing scheduling and runtime techniques to exploit fine- and coarse-grained parallelism in collaboration with Prof. Yatin Manerkar.

### Carnegie Mellon University

Pittsburgh, PA

GRADUATE RESEARCH ASSISTANT

June 2023 - May 2026

- Researching in computer architecture and performance modeling.

## AMD Research

RESEARCH INTERN

San Jose, CA

June 2024 - August 2024

- Implemented and profiled applications on near-cache accelerators.

## Carnegie Mellon University

UNDERGRADUATE RESEARCH ASSISTANT

Pittsburgh, PA

May 2022 - May 2023

- Designed and verified cache-coherence protocols for near-cache accelerators. Evaluated performance using architecture simulation.

## University of Portland

UNDERGRADUATE RESEARCHER

Portland, OR

Jan. 2022 - May 2022

- Evaluated CPU specialization for Viterbi decoding using architecture simulation, power modeling, and low-level software optimization.

## Intelligent, Complex, Adaptive, and Networks Lab

UNDERGRADUATE RESEARCH ASSISTANT

University of Portland

May 2021 - August 2021

- Developed software for conducting EEG experiments. Built mechanisms to coordinate and synchronize experimental components.

## Service & Leadership

### Doctoral Review Committee

GRADUATE STUDENT REPRESENTATIVE

Carnegie Mellon University CSD

Spring 2025 - Spring 2026

### Graduate Student Assembly

COMPUTER SCIENCE DEPARTMENT REPRESENTATIVE

Carnegie Mellon University

Fall 2023 - Fall 2024

### Tau Beta Pi

OREGON GAMMA CHAPTER PRESIDENT

University of Portland

2021 - 2022

## Mentoring

### Research Advising

Amin Hong (CMU ECE master's student)

Summer 2025

Bas Yoovidhya (CMU CS master's (thesis) student)

Fall 2023 - Summer 2024

Mayne Mei (University of Michigan CS undergraduate, co-advised with Prof. Yatin Manerkar)

Fall 2023

## Teaching

### Carnegie Mellon University

Introductory Course in Leadership and Professionalism (IC-LEAP)

Instructor, Spring 2025

### University of Portland

Theory of Computation (CS 357)

Grader, Fall 2022

Electrical Engineering Dept. Tutor (EE 231, 261, 262, & 332)

Tutor, 2021-2022

Logic Design (EE 231)

Grader, Fall 2021

Electrical Circuits Lab (EE 271)

Lab Assistant, Spring 2021

## Skills

### Programming

C/C++, Python, Java, CUDA, oneTBB, pthreads, RISC-V Assembly, MATLAB, some Haskell and Verilog

### Systems & Compilers

Linux/Unix, LLVM, gem5, Intel Pin, Linux perf, Murphi Model Checker, Nsight Systems & Compute